

MOHAMMAD AARISH KHAN

Biomedical Engineering Major
Honors College
University at Buffalo

EDUCATION

University at Buffalo
Honors College
Fall 2026 – Present

Williamsville North High School
Class of 2026
SAT: 1510
AP Scholar of Distinction, 2025

TECHNICAL SKILLS

- **Programming:** Python, Java, Arduino, R
- **AI/ML:** AlphaFold, PyTorch, Hugging Face, Crew AI, Ollama
- **Comp. Biology:** GROMACS, PLIP, Prodigy, HDock, FastENLOC, Biopython, Benchling
- **Web Dev:** Streamlit, Google Colab, Notion
- **Cloud:** Oracle Cloud Infrastructure (Gen AI Certified)

AWARDS & HONORS

- **NY Academy of Sciences:** Emerging Student Researcher of the Year 2025 (Top 5 of ~8,000 internationally)
- 1st Place 2023; Project of Distinction 2024
- **ACS NERD:** Outstanding Poster Presentation Award 2025
- **RIT:** Science and Math Award 2025 (\$24,000/yr scholarship)
- **Terra Science Fair:** Medal of High Honors; Office of Naval Research Award
- **TSA NY State:** 1st Place, T-Shirt Design 2025; 3rd Place, Promotional Graphics 2025 & Coding 2024
- **College Board:** AP Scholar of Distinction 2025
- **Science Olympiad Regionals:** 2nd Place, Disease Detective 2023
- **CS4HS, Buffalo State:** 1st Place, Computer Programming 2024
- **President's Volunteer Service Award** 2025 (Bronze, 600+ hrs)

WORKSHOPS & CERTS

- MIT Beaver Works Summer Institute — Autonomous Cars Course
- Biotech in Action — Biogen Community Lab & Lemelson-MIT
- Oracle Cloud Infrastructure 2024 Gen AI Certification
- Eric Pitman Summer Workshop 2025 — Intern Computational Biologist

CONFERENCES

- Kennedy Space Center 2025 (NASA SmallSat)
- iFORE 2025
- ACS NERD 2025
- Terra Science Fair
- Global Impact Council Conference
- IEEE ISEC 2025

RESEARCH EXPERIENCE

CURRENT RESEARCH

Williams Lab — ADHD Genetics Research *University at Buffalo*

- Use the **FastENLOC** computational pipeline to analyze genetic data as part of an ongoing research study investigating the genetic basis of ADHD

Nushoor — Water Purification Research *Local Incubator for Underserved Communities*

- Collaborate on a team designing a low-cost water purification system for refugee communities in Bangladesh
- Tailor system design to community-specific constraints, including limited resources, cost, maintenance, and environmental conditions

PAST RESEARCH

Bispecific Antibodies for Cancer Immunotherapy *First Author, Published Researcher — Eigen Sciences*

- Designed a bispecific DuoBody antibody targeting 4-1BB on cancer cells and PD-L1 on T-cells as a potential treatment for non-small cell lung cancer (NSCLC), working with 10 antibodies, 2 receptors, and 6+ computational platforms (AlphaFold, GROMACS, HDock, PLIP, Prodigy)
- Published “Bispecific DuoBody Antibody Targeting 4-1BB on Cancer Cells and PD-L1 on T-Cells for Immunotherapy in Non-Small Cell Lung Cancer (NSCLC)” — IEEE ISEC 2025 (DOI: 10.1109/ISEC64801.2025.11147433)
- Created two research posters and presented at five conferences: ACS NERD, iFORE, Terra Science Fair, Global Impact Council Conference, and IEEE ISEC
- Recognition: **Outstanding Poster Presentation Award** (ACS NERD 2025); **Medal of High Honors** and **Office of Naval Research Award** (Terra Science Fair)

DuoDok — Independent Computational Research Tool *Personal Project*

- Built a simulated pipeline automating portions of the bispecific antibody computational workflow, deployed as a web application
- Authored a research abstract and presented at iFORE 2025 and ACS NERD 2025, with an upcoming conference presentation planned

High-Altitude Balloon Research *UB Nanosatellite Lab — Summer Internship*

- Designed electronics and wrote code as part of a 5-student team for a high-altitude balloon mission carrying six solar panels
- Studied variation in solar panel efficiency across altitude; presented findings via PowerPoint and research abstract at the SmallSat Education Conference, Kennedy Space Center (NASA)

International Research Projects *Head of Ideation & Implementation — NY Academy of Sciences Junior Academy Challenge*

- Led ideation, planning, and solution development for two international teams: an asteroid mining project (**1st Place**) and a multiple sclerosis health wearable (**Project of Distinction**)
- Co-authored abstracts and PowerPoint presentations for both projects
- Selected as 1 of 5 students internationally, out of roughly 8,000 applicants, for the **Emerging Student Researcher of the Year** award

Environmental Genomics *Eric Pitman Summer Workshop, University at Buffalo*

- Analyzed water samples from Scajaquada Creek using whole-genome analysis and R-based programming to characterize microorganisms and biological material
- Presented findings in a final research presentation; gained hands-on experience in genomics, bioinformatics, R programming, and data analysis

LEADERSHIP & EXTRACURRICULARS

- **Computer Club** — President (2025–Present), Vice President (2023–25)
- **National Honors Society** — Vice President, Class of 2026 (2025–Present); organized NHS Running Event and co-organized Tech Mentorship Program for middle schoolers
- **Technology Student Association** — Secretary & NY State Delegate (2024–25)
- **Global Impact Council** — Director of Social Media (2024–25); Writing Team (2023–24); Co-Organizer, Emerging Research Incubator
- **Superintendent Inter-High Action Committee** — Co-Chair (2022–Present)
- **Shared Decision Making Committee** — Class Representative (2024–Present)
- **Science Olympiad** (2022–Present) | **Science Honors Society** (2023–Present)